

Point and Variance Estimators Implemented in **sreg**

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1 Notation

Object	Meaning
$d \in \mathcal{D} \setminus \{0\}$	focal treatment arm; 0 is control
$i = 1, \dots, N$	individual; outcome Y_i , treatment D_i , stratum S_i , adjustment covariates X_i
$g = 1, \dots, G$	cluster; size N_g , treatment D_g , stratum S_g , covariates X_g
T_g	expanded cluster outcome, $T_g = N_g \bar{Y}_g$
$N_r(s), G_r(s)$	counts assigned to arm r in stratum s ; $N(s), G(s)$ are stratum totals
$\hat{\pi}_r(s)$	$N_r(s)/N(s)$ for individuals; $G_r(s)/G(s)$ for clusters
λ_s	units in small stratum s ; $ \lambda_s = k$, with $k(r)$ assigned to arm r and $\pi_r = k(r)/k$
$\bar{N}_G$	$G^{-1} \sum_g N_g$; N_c, G_c denote mixed-component totals, $c \in \{s, \ell\}$

All formulas are evaluated separately for each $d \in \mathcal{D} \setminus \{0\}$. For multiple treatments, $\hat{\pi}_d(s)$ and $\hat{\pi}_0(s)$ are used separately. Setting the fitted adjustment terms equal to zero gives the unadjusted estimators.

2 Large-Strata Estimators

2.1 Individual-Level Treatment Assignment

For $r \in \mathcal{D}$ and $s \in \mathcal{S}$, fit the treatment-by-stratum regression and retain its slope:

$$Y_i = \gamma_{r,s} + X_i^\top \theta_{r,s} + u_i, \quad \hat{\mu}(r, s, X_i) = X_i^\top \hat{\theta}_{r,s}.$$

Define

$$\begin{aligned} \widehat{M}_{d,i} &= \hat{\mu}(d, S_i, X_i) - \hat{\mu}(0, S_i, X_i), \\ \hat{\psi}_{d,i}^{\text{large,adj}} &= \frac{\mathbf{1}\{D_i = d\}\{Y_i - \hat{\mu}(d, S_i, X_i)\}}{\hat{\pi}_d(S_i)} - \frac{\mathbf{1}\{D_i = 0\}\{Y_i - \hat{\mu}(0, S_i, X_i)\}}{\hat{\pi}_0(S_i)} + \widehat{M}_{d,i}. \end{aligned}$$

The point estimator is

$$\boxed{\hat{\tau}_N^{\text{large,adj}}(d) = \frac{1}{N} \sum_{i=1}^N \hat{\psi}_{d,i}^{\text{large,adj}}} \quad \left[\hat{\mu} \equiv 0 \implies \hat{\tau}_N^{\text{large}}(d) = \frac{1}{N} \sum_i \left\{ \frac{\mathbf{1}\{D_i = d\}Y_i}{\hat{\pi}_d(S_i)} - \frac{\mathbf{1}\{D_i = 0\}Y_i}{\hat{\pi}_0(S_i)} \right\} \right].$$

For the variance estimator, set

$$\begin{aligned}
\widehat{\Xi}_{d,i} &= \widehat{M}_{d,i} + \frac{Y_i - \widehat{\mu}(d, S_i, X_i)}{\widehat{\pi}_d(S_i)}, \quad \widehat{\Xi}_{0,i} = \widehat{M}_{d,i} - \frac{Y_i - \widehat{\mu}(0, S_i, X_i)}{\widehat{\pi}_0(S_i)}, \quad \widehat{\Xi}_{-d,0,i} = \widehat{M}_{d,i}, \\
\widetilde{\Xi}_r(s) &= \frac{1}{N_r(s)} \sum_{j:D_j=r, S_j=s} \widehat{\Xi}_{r,j} \quad (r \in \{d, 0\}), \quad \overline{M}_d(s) = \frac{1}{N(s)} \sum_{j:S_j=s} \widehat{M}_{d,j}, \\
\widehat{\Xi}_{d,i} &= \widehat{\Xi}_{d,i} - \widetilde{\Xi}_d(S_i), \quad \widehat{\Xi}_{0,i} = \widehat{\Xi}_{0,i} - \widetilde{\Xi}_0(S_i), \quad \widehat{\Xi}_{-d,0,i} = \widehat{M}_{d,i} - \overline{M}_d(S_i), \\
\widehat{\tau}_{N,s}(d) &= \frac{1}{N_d(s)} \sum_{j:D_j=d, S_j=s} Y_j - \frac{1}{N_0(s)} \sum_{j:D_j=0, S_j=s} Y_j, \quad \widehat{\Xi}_2(s) = \widehat{\tau}_{N,s}(d) - \widehat{\tau}_N^{\text{large,adj}}(d).
\end{aligned}$$

Then

$$\widehat{V}_N^{\text{large,adj}}(d) = \frac{1}{N} \sum_{i=1}^N \left[\mathbf{1}\{D_i = d\} \widehat{\Xi}_{d,i}^2 + \mathbf{1}\{D_i = 0\} \widehat{\Xi}_{0,i}^2 + \mathbf{1}\{D_i \notin \{d, 0\}\} \widehat{\Xi}_{-d,0,i}^2 + \widehat{\Xi}_2(S_i)^2 \right].$$

2.2 Cluster-Level Treatment Assignment

For $r \in \mathcal{D}$ and $s \in \mathcal{S}$, fit

$$T_g = \gamma_{r,s} + X_g^\top \theta_{r,s} + u_g, \quad \widehat{\mu}(r, s, X_g) = X_g^\top \widehat{\theta}_{r,s}, \quad g \in \{g : D_g = r, S_g = s\}.$$

With $\widehat{M}_{d,g} = \widehat{\mu}(d, S_g, X_g) - \widehat{\mu}(0, S_g, X_g)$, define

$$\widehat{\Xi}_g^{\text{cl,large}}(d) = \frac{\mathbf{1}\{D_g = d\} \{T_g - \widehat{\mu}(d, S_g, X_g)\}}{\widehat{\pi}_d(S_g)} - \frac{\mathbf{1}\{D_g = 0\} \{T_g - \widehat{\mu}(0, S_g, X_g)\}}{\widehat{\pi}_0(S_g)} + \widehat{M}_{d,g}.$$

The ratio estimator is

$$\widehat{\tau}_G^{\text{cl,large}}(d) = \frac{\sum_{g=1}^G \widehat{\Xi}_g^{\text{cl,large}}(d)}{\sum_{g=1}^G N_g}.$$

For the variance estimator, define

$$\begin{aligned}
\overline{N}_G(s) &= \frac{1}{G(s)} \sum_{j:S_j=s} N_j, \quad R_g(d) = \widehat{\tau}_G^{\text{cl,large}}(d) \{N_g - \overline{N}_G(S_g)\}, \\
\widehat{\Xi}_{d,g} &= \widehat{M}_{d,g} + \frac{T_g - \widehat{\mu}(d, S_g, X_g)}{\widehat{\pi}_d(S_g)}, \quad \widehat{\Xi}_{0,g} = \widehat{M}_{d,g} - \frac{T_g - \widehat{\mu}(0, S_g, X_g)}{\widehat{\pi}_0(S_g)}, \quad \widehat{\Xi}_{-d,0,g} = \widehat{M}_{d,g}, \\
\widetilde{\Xi}_r(s) &= \frac{1}{G_r(s)} \sum_{j:D_j=r, S_j=s} \widehat{\Xi}_{r,j} \quad (r \in \{d, 0\}), \quad \overline{M}_d(s) = \frac{1}{G(s)} \sum_{j:S_j=s} \widehat{M}_{d,j}, \\
\widehat{\Xi}_{d,g} &= \widehat{\Xi}_{d,g} - \widetilde{\Xi}_d(S_g) - R_g(d), \quad \widehat{\Xi}_{0,g} = \widehat{\Xi}_{0,g} - \widetilde{\Xi}_0(S_g) - R_g(d), \\
\widehat{\Xi}_{-d,0,g} &= \widehat{M}_{d,g} - \overline{M}_d(S_g) - R_g(d), \\
\widehat{\Xi}_2(s) &= \frac{1}{G_d(s)} \sum_{j:D_j=d, S_j=s} T_j - \frac{1}{G_0(s)} \sum_{j:D_j=0, S_j=s} T_j - \widehat{\tau}_G^{\text{cl,large}}(d) \overline{N}_G(s).
\end{aligned}$$

Then

$$\widehat{V}_G^{\text{cl,large}}(d) = \frac{1}{N_G^2} \frac{1}{G} \sum_{g=1}^G \left[\mathbf{1}\{D_g = d\} \widehat{\Xi}_{d,g}^2 + \mathbf{1}\{D_g = 0\} \widehat{\Xi}_{0,g}^2 + \mathbf{1}\{D_g \notin \{d, 0\}\} \widehat{\Xi}_{-d,0,g}^2 + \widehat{\Xi}_2(S_g)^2 \right].$$

3 Small-Strata Estimators

3.1 Individual-Level Treatment Assignment

For $s = 1, \dots, n$, form the within-stratum contrasts

$$\begin{aligned} \widehat{\delta}_{Y,s}(d) &= \frac{1}{k(d)} \sum_{i \in \lambda_s} Y_i \mathbf{1}\{D_i = d\} - \frac{1}{k(0)} \sum_{i \in \lambda_s} Y_i \mathbf{1}\{D_i = 0\}, \\ \widehat{\delta}_{X,s}(d) &= \frac{1}{k(d)} \sum_{i \in \lambda_s} X_i \mathbf{1}\{D_i = d\} - \frac{1}{k(0)} \sum_{i \in \lambda_s} X_i \mathbf{1}\{D_i = 0\}. \end{aligned}$$

Obtain $(\widehat{\gamma}_d, \widehat{\beta}_d)$ from Group OLS, $\widehat{\delta}_{Y,s}(d) = \gamma_d + \widehat{\delta}_{X,s}(d)^\top \beta_d + v_s$, and let $\overline{X}_N = N^{-1} \sum_i X_i$. Then

$$\begin{aligned} \widehat{\tau}_N^{\text{small,adj}}(d) &= \frac{1}{N_d} \sum_i Y_i \mathbf{1}\{D_i = d\} - \frac{1}{N_0} \sum_i Y_i \mathbf{1}\{D_i = 0\} \\ &\quad - \left[\frac{1}{N_d} \sum_i (X_i - \overline{X}_N) \mathbf{1}\{D_i = d\} - \frac{1}{N_0} \sum_i (X_i - \overline{X}_N) \mathbf{1}\{D_i = 0\} \right]^\top \widehat{\beta}_d = \widehat{\gamma}_d. \end{aligned}$$

Define $\widehat{Y}_i^a = Y_i - (X_i - \overline{X}_N)^\top \widehat{\beta}_d$. For $r, r' \in \{d, 0\}$, let

$$\begin{aligned} \widehat{\Gamma}_N(r) &= \frac{1}{nk(r)} \sum_i \widehat{Y}_i^a \mathbf{1}\{D_i = r\}, \quad \widehat{\sigma}_N^2(r) = \frac{1}{nk(r)} \sum_i (\widehat{Y}_i^a - \widehat{\Gamma}_N(r))^2 \mathbf{1}\{D_i = r\}, \\ \widehat{\rho}_N(r, r') &= \begin{cases} \frac{1}{n} \sum_{s=1}^n \frac{1}{k(r)k(r')} \left(\sum_{i \in \lambda_s} \widehat{Y}_i^a \mathbf{1}\{D_i = r\} \right) \left(\sum_{i \in \lambda_s} \widehat{Y}_i^a \mathbf{1}\{D_i = r'\} \right), & r \neq r', \\ \frac{2}{n} \sum_{j=1}^{n/2} \frac{1}{k(r)^2} \left(\sum_{i \in \lambda_{2j-1}} \widehat{Y}_i^a \mathbf{1}\{D_i = r\} \right) \left(\sum_{i \in \lambda_{2j}} \widehat{Y}_i^a \mathbf{1}\{D_i = r\} \right), & r = r'. \end{cases} \end{aligned}$$

Set

$$\widehat{V}_{2,N}(r, r') = \widehat{\rho}_N(r, r') - \widehat{\Gamma}_N(r) \widehat{\Gamma}_N(r'), \quad \widehat{V}_{1,N}(r) = \widehat{\sigma}_N^2(r) - \widehat{V}_{2,N}(r, r).$$

The plug-in variance estimator is

$$\widehat{V}_N^{\text{small,adj}}(d) = \frac{1}{\pi_d} \widehat{V}_{1,N}(d) + \frac{1}{\pi_0} \widehat{V}_{1,N}(0) + \widehat{V}_{2,N}(d, d) + \widehat{V}_{2,N}(0, 0) - 2\widehat{V}_{2,N}(d, 0).$$

3.2 Cluster-Level Treatment Assignment

For $s = 1, \dots, n$, form

$$\begin{aligned}\widehat{\delta}_{T,s}^{\text{cl}}(d) &= \frac{1}{k(d)} \sum_{g \in \lambda_s^{\text{cl}}} T_g \mathbf{1}\{D_g = d\} - \frac{1}{k(0)} \sum_{g \in \lambda_s^{\text{cl}}} T_g \mathbf{1}\{D_g = 0\}, \\ \widehat{\delta}_{X,s}^{\text{cl}}(d) &= \frac{1}{k(d)} \sum_{g \in \lambda_s^{\text{cl}}} X_g \mathbf{1}\{D_g = d\} - \frac{1}{k(0)} \sum_{g \in \lambda_s^{\text{cl}}} X_g \mathbf{1}\{D_g = 0\}.\end{aligned}$$

Obtain $(\widehat{\gamma}_d^{\text{cl}}, \widehat{\beta}_d^{\text{cl}})$ from

$$\widehat{\delta}_{T,s}^{\text{cl}}(d) = \gamma_d^{\text{cl}} + \widehat{\delta}_{X,s}^{\text{cl}}(d)^\top \beta_d^{\text{cl}} + v_s.$$

Then

$$\widehat{\tau}_G^{\text{cl,small}}(d) = \frac{\widehat{\gamma}_d^{\text{cl}}}{\overline{N}_G}.$$

Let $\overline{X}_G = G^{-1} \sum_g X_g$ and define

$$\widehat{R}_g^a = T_g - (X_g - \overline{X}_G)^\top \widehat{\beta}_d^{\text{cl}}, \quad \widehat{W}_g^a(r) = \frac{1}{\overline{N}_G} \left[\{\mathbf{1}\{r = d\} - \mathbf{1}\{r = 0\}\} \widehat{R}_g^a - \pi_r \widehat{\tau}_G^{\text{cl,small}}(d) N_g \right], \quad r \in \mathcal{D}.$$

For $r, r' \in \mathcal{D}$, set

$$\begin{aligned}\widehat{\Gamma}_G(r) &= \frac{1}{nk(r)} \sum_g \widehat{W}_g^a(r) \mathbf{1}\{D_g = r\}, \quad \widehat{\sigma}_G^2(r) = \frac{1}{nk(r)} \sum_g (\widehat{W}_g^a(r) - \widehat{\Gamma}_G(r))^2 \mathbf{1}\{D_g = r\}, \\ \widehat{\rho}_G(r, r') &= \begin{cases} \frac{1}{n} \sum_{s=1}^n \frac{1}{k(r)k(r')} \left(\sum_{g \in \lambda_s^{\text{cl}}} \widehat{W}_g^a(r) \mathbf{1}\{D_g = r\} \right) \left(\sum_{g \in \lambda_s^{\text{cl}}} \widehat{W}_g^a(r') \mathbf{1}\{D_g = r'\} \right), & r \neq r', \\ \frac{2}{n} \sum_{j=1}^{n/2} \frac{1}{k(r)^2} \left(\sum_{g \in \lambda_{2j-1}^{\text{cl}}} \widehat{W}_g^a(r) \mathbf{1}\{D_g = r\} \right) \left(\sum_{g \in \lambda_{2j}^{\text{cl}}} \widehat{W}_g^a(r) \mathbf{1}\{D_g = r\} \right), & r = r'. \end{cases}\end{aligned}$$

Set

$$\widehat{V}_{2,G}(r, r') = \widehat{\rho}_G(r, r') - \widehat{\Gamma}_G(r) \widehat{\Gamma}_G(r'), \quad \widehat{V}_{1,G}(r) = \widehat{\sigma}_G^2(r) - \widehat{V}_{2,G}(r, r).$$

The plug-in variance estimator is

$$\widehat{V}_G^{\text{cl,small}}(d) = \sum_{r \in \mathcal{D}} \frac{1}{\pi_r} \widehat{V}_{1,G}(r) + \sum_{r \in \mathcal{D}} \sum_{r' \in \mathcal{D}} \widehat{V}_{2,G}(r, r').$$

4 Mixed-Design Estimators

4.1 Individual-Level Treatment Assignment

Partition observations into small- and large-strata components of sizes N_s and N_ℓ , $N = N_s + N_\ell$. Then

$$\widehat{\tau}_N^{\text{mixed}}(d) = \frac{N_s}{N} \widehat{\tau}_{N_s}^{\text{small}}(d) + \frac{N_\ell}{N} \widehat{\tau}_{N_\ell}^{\text{large}}(d).$$

$$\begin{aligned}\widehat{V}_N^{\text{mixed}}(d) &= \frac{N_s}{N} \widehat{V}_{N_s}^{\text{small}}(d) + \frac{N_\ell}{N} \widehat{V}_{N_\ell}^{\text{large}}(d) \\ &\quad + \frac{N_s N_\ell}{N^2} \left\{ \widehat{\tau}_{N_\ell}^{\text{large}}(d) - \widehat{\tau}_{N_s}^{\text{small}}(d) \right\}^2.\end{aligned}$$

4.2 Cluster-Level Treatment Assignment

Let G_c and $N_c = \sum_{g \in \mathcal{G}_c} N_g$ be the cluster and individual totals in component $c \in \{s, \ell\}$. Define

$$\widehat{q}_c = \frac{G_c}{G}, \quad \widehat{p}_c = \frac{N_c}{N}, \quad \widehat{V}_{p,G} = \frac{1}{G N_G^2} \sum_{g=1}^G N_g^2 \{ \mathbf{1}\{g \in \mathcal{G}_\ell\} - \widehat{p}_\ell \}^2.$$

Then

$$\begin{aligned}\widehat{\tau}_G^{\text{cl,mixed}}(d) &= \frac{N_s}{N} \widehat{\tau}_{G_s}^{\text{cl,small}}(d) + \frac{N_\ell}{N} \widehat{\tau}_{G_\ell}^{\text{cl,large}}(d). \\ \widehat{V}_G^{\text{cl,mixed}}(d) &= \frac{\widehat{p}_s^2}{\widehat{q}_s} \widehat{V}_{G_s}^{\text{cl,small}}(d) + \frac{\widehat{p}_\ell^2}{\widehat{q}_\ell} \widehat{V}_{G_\ell}^{\text{cl,large}}(d) \\ &\quad + \widehat{V}_{p,G} \left\{ \widehat{\tau}_{G_\ell}^{\text{cl,large}}(d) - \widehat{\tau}_{G_s}^{\text{cl,small}}(d) \right\}^2.\end{aligned}$$

5 Standard Errors

$$\widehat{\text{se}}(\widehat{\tau}_N(d)) = \sqrt{\frac{\widehat{V}_N(d)}{N}}, \quad \widehat{\text{se}}(\widehat{\tau}_G^{\text{cl}}(d)) = \sqrt{\frac{\widehat{V}_G^{\text{cl}}(d)}{G}}.$$